scientific reports

OPEN

Dynamics of Eucalyptus and Sorghum biomass growth and nitrogen assessment at a Saharan sandy soil irrigated with treated wastewater

Aicha Serraye[1]✉, Tahar Idder[1], Mustapha Daddi Bouhoun[2], Jean B. Diatta[3], Ahmed Tani[4], Paul Genz[5], Marcel Ackermann[6] & Mohammed S. Nili[7]

The study outlines the dynamics of Eucalyptus and Sorghum biomass under treated wastewater (TWW) irrigation in the Algerian Sahara to encourage water recycling and green biomass production regarding the freshwater (FW) shortage and plant cover rarity. Eucalyptus samples were taken at 03 different ages and 03 distinct layers. Sorghum samples were conducted from 2018 to 2020. The results proved a significant growth in Eucalyptus branch biomass and total biomass carbon of Eucalyptus branches in the layers 0–1 m and 2–4 m at 3 and 7 years old trees due to TWW irrigation. Eucalyptus aboveground biomass increased more with FW irrigation (mean values were 204.56 kg and 364.30 kg) compared to TWW irrigation (mean values were 126.64 kg and 222.56 kg) for 7 and 10 years old trees, respectively, across the ascending tree ages. Total nitrogen in Eucalyptus leaves was positively affected by TWW in the layers 2–4 m at 7 years old trees, with mean values of 3.10%. For Sorghum plants, TWW had a significant positive impact on the aboveground biomass, measuring 1.08 kg/m^2 in autumn and 1.13 kg/m^2 in spring, likewise on the total carbon biomass and nitrogen content. TWW irrigation can be an option for biomass growth for the studied vegetation in deserts.

Keywords Eucalyptus, Sorghum, Biomass dynamics, Treated wastewater, Nitrogen content, Saharan region

The existence on earth rests on water due to its necessity in building the life of living beings. Its quantity can cause conflict on a global scale[1,2]. In 2025, the global population is projected to reach 9 billion people[3], with 1.8 billion people facing water shortages[4]. Where serious cases were detected in North Africa, North China, Northwest India, southwestern and central parts of the United States, Pakistan, and Mexico[5].

As a study case, Algeria contains the hottest and driest part of the Sahara[6,7]. It is classified as the second-most water-scarce African country, with a rainfall shortage of approximately 30% in the previous 30 years. The freshwater (FW) availability was 430 m^3/capita/year in 2020[8], while the theoretical scarcity threshold set by the World Bank is 1000 m^3/capita/year[9]. For that reason, Algeria is listed as a nation with deficient water resources. Currently, Algeria is profoundly vulnerable to climate change[10]. This Mediterranean country is facing the menace of desertification and land degradation, particularly in the southern part, where drought and lack of precipitation extend for long periods[11]. Therefore, meeting the increasing demand for water is a primary concern worldwide[12]. Hence, the management of alternative water resources is a highly acute universal issue especially in arid areas[13].

[1]Laboratory of Phoeniciculture Research, Department of Biology, University of Kasdi Merbah, BP 511 Route de Ghardaïa, 30000 Ouargla, Algeria. [2]Laboratory of Ecosystem Protection in Arid and Semi-Arid Zones, Department of Agricultural Sciences, University of Kasdi Merbah, B.P. 511 Route de Ghardaïa, 30000 Ouargla, Algeria. [3]Department of Agricultural Chemistry and Environmental Biogeochemistry, Poznan University of Life Sciences, ul. Wojska Polskiego 71F, 60-625 Poznan, Poland. [4]Laboratory of Quality Control and Fraud Prevention of El-Oued, District of 8th May 1945, 39001 El-Oued, Algeria. [5]Department Analytics, Helmholtz Centre for Environmental Research – UFZ, Permoserstr. 15, 04318 Leipzig, Germany. [6]Faculty of Agricultural and Environmental Sciences, University of Rostock, Professorship Crop Production, Justus-Von-Liebig-Weg 6, 18059 Rostock, Germany. [7]Laboratory of Biodiversity and Biotechnology Applications in Agriculture, University of El-Oued, 39000 El-Oued, Algeria. ✉email: serraye-aicha@univ-eloued.dz

The principal goal of introducing Eucalyptus in public places and roadsides in the Algerian desert cities and using Sorghum as a greenbelt in the farm borders is to boost the air quality. Sorghum can improve air quality by reducing dust storms and stabilizing the movement of sand dunes while attempting to sustain land from strong wind erosion and desertification. However, these measures increase the irrigation needs in this drought area[14,15].

Algerian authorities are seriously focusing on increasing water resources throughout the territory by expanding the projects of reusing treated wastewater (TWW) in most parts of the state[16]. Only 8% of this alternative water source is reused in some Algerian *wilayas* (administrative divisions in certain African and Asian countries)[17]. According to a national report[18], 400 million m^3 of wastewater is treated per year across 199 wastewater treatment plants (WWTPs) in Algeria. This considerable volume is now seen as one of the biggest strategic solutions for water scarcity[19], especially in the Saharan part of the country.

Likewise, FW is consumed by Sorghum in the Algerian desert[20], where farmers cultivate this plant to feed livestock and to act as a windbreak. Many studies have shown that this cereal tolerates heat, drought stress, and a dry climate. It develops well in sandy soils under regular irrigation and with low agrochemical inputs[21–26]. It can reach the highest levels of productivity in arid environments due to the high temperatures[27]. Furthermore, this C4 crop takes up high amounts of nitrogen (N) to build dry matter (DM)[22].

Besides, 90% of plant biomass is mainly composed of carbon (C), oxygen (O), hydrogen (H)[28,29], while 50% of the woody tree biomass is C[30]. Hence, building plant biomass assimilates the greenhouse gas carbon dioxide (CO_2) from the atmosphere and sequesters carbon in plants and soil, which improves the soil of marginal lands and mitigates climate change[31,32]. The foliar N also contributes to the growth of the aboveground trees and biomass production via photosynthesis. Its sequestration differs depending on the kind of tree[33] and the storage of this element in the environment compartments[34–36].

In southern Algeria, in addition to municipal and industrial exploitation purposes, FW is used to irrigate plants[37] such as Eucalyptus[38,39]. Eucalyptus adapts well to aridity conditions and can grow on fragile soils when water is available[40]. It has been found in Australia and belongs to the Myrtaceae family. There are about 700 species across different ecosystems, including saline environments, and Eucalyptus has a high economic value[41]. This tree was introduced to Algeria between 1854 and 1860[42]. In semi-arid regions, it is utilized for afforestation as it increases the concentrations of total nitrogen, phosphorus, and calcium in degraded soils[43]. It demands a lot of water—up to 90 L/day. It consumes up to 785 L to produce 1 kg of biomass. Eucalyptus can sequestrate carbon into the soil reservoir[44,45], which helps fight desertification in arid regions[46,47].

TWW has various positive effects on Eucalyptus and Sorghum plants[41,48]. For example, TWW contains many nutrients that can improve the development of leaves, and it has ramifications for the number, diameter, and height of Eucalyptus trees through irrigation processes[49].

In addition to its considerable effects on plants, TWW contributes to carbon sequestration in sandy soils following good irrigation management[50]. Its usage conserves drinking water and helps balance water supply and water demand[51]. Irrigation with TWW can fight climate change by decreasing greenhouse gas emissions resulting from traditional agricultural fertigation[52]. Moreover, the reuse of this alternative source promotes the circulation of water and nutrients in the environmental compartments instead of contaminating natural water bodies and posing an environmental risk[53,54]. However, negative effects can occur under uncontrolled irrigation, which contributes to a chemical risk depending on the quality of the treated water. As a result, the soil can be contaminated by directly receiving and accumulating these chemicals, leading to phytotoxicity[55–57].

As far as we know, no existing scientific reports have investigated the path of biomass in different plant heights or nitrogen evaluation in this vegetation under TWW irrigation in the dunes area of Algeria. Thus, our objectives presented in this paper are as follows:

(1) To evaluate the usefulness and productivity effect of TWW in the desert environment.
(2) To study the dynamics of the biomass especially the distribution of the carbon biomass and nitrogen content in the plants under TWW irrigation in the desert environment.

Materials and methods

Study area and plant sampling

The study area is El-Oued City, located in the Eastern South of the Algerian Sahara. It has a total area of 11,738.4 km^2[58] and is situated at latitude 33°27′20″ N and longitude 7°11′ 0″ E. El-Oued is characterized by two geological formations: continuous dunes from the Quaternary formations and the Great Eastern Sand Sea. The region is known for its excessively high sunshine duration, limited vegetation, dispersed oases, and salt lakes[59]. The climate in El-Oued is hyper-arid[60], with an annual evaporation rate of over 2200 mm and extremely hot temperatures in summer exceeding 45 °C. The annual precipitation is around 65 mm, and the average annual humidity is approximately 47.39%[58,61]. Due to these conditions, plants can only receive sufficient water through irrigation[62].

The research sites were selected on the basis of the originality of the study and the availability of two types of irrigation water. Site 1 and site 3 were located at the WWTP of Kouinine municipality, where urban TWW is accessible, with permission granted solely for scientific investigations. These sites had TWW irrigation. The TWW was obtained from a discharge water pipe at Kouinine WWTP. This water was filtered before the irrigation in the experimental plots, and the filters were changed weekly. According to the national sanitation office in El-Oued, the TWW is treated using an aerated lagoon system, and all the treated water is transferred daily to the Chott Halloufa outlet without any agreement for reuse. Near the WWTP of Kouinine, Sites 2 and 4 were installed and received FW irrigation from a well drilled to a depth of around 20 m. These sites were selected for their soil and climate conditions similar to those of sites 1 and 3 (Fig. 1). The soil of the experimental sites is sandy. It consists of around 97.63% quartz, 0.56% minor calcite mineral, and very low concentrations of other oxides. The bulk density is 1.6 g/mL, and the porosity measures 34.09%[59]. The soil's structure is characterized by very

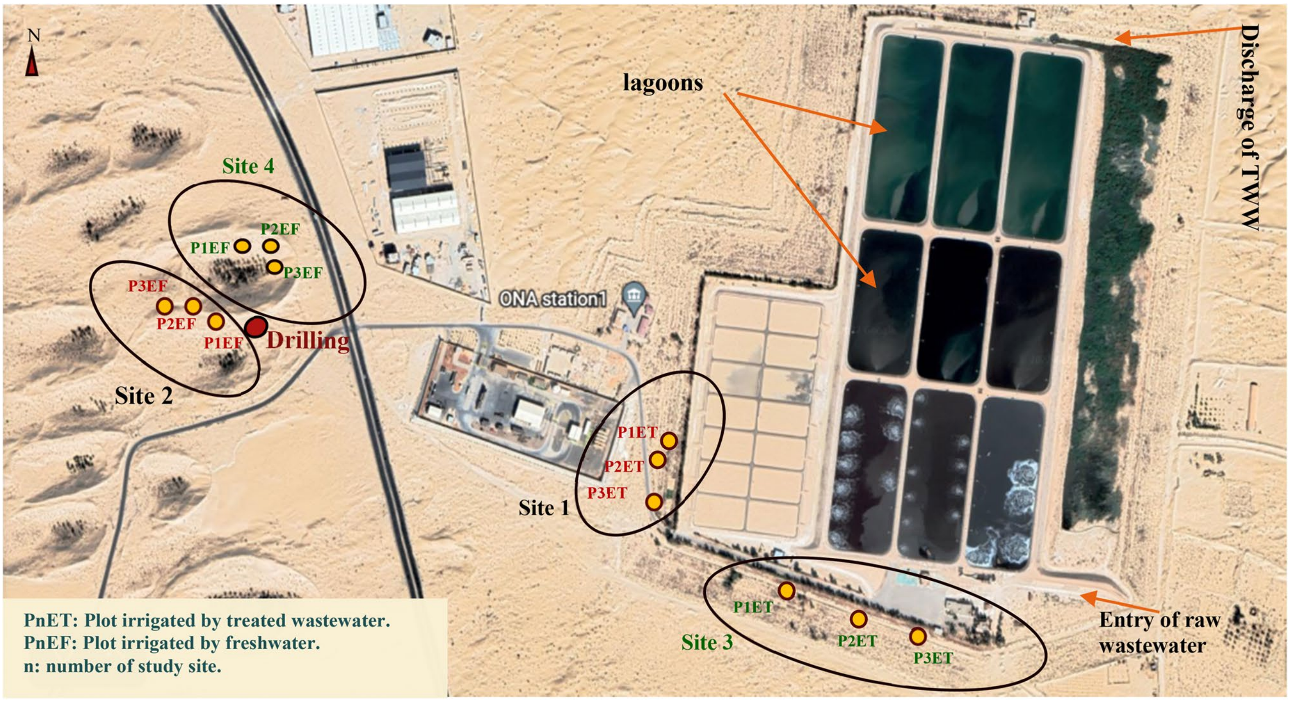

Fig. 1. Location of the study sites and experimental plots (Map source: Google Earth on web, 33°25′09″N 6°51′04″E, elevation 72 m. URL link: https://earth.google.com/web Images©2020 Airbus, CNES/Airbus, Maxar Technologies, Map Data©2020.)

high water permeability and very low organic matter content[63]. It did not get any compost during the experiment period and had not been cultivated before the setup of the study.

In this study area, site 1 and site 2 were selected for the planting of Sorghum. Site 3 and site 4 were chosen for Eucalyptus investigation as shown in Fig. 1. Each site was divided into three plots regarding the sampling method. The plots irrigated with TWW were: P1ET, P2ET, and P3ET. The plots irrigated with FW were: P1EF, P2EF, and P3EF. In total, there were 12 experimental plots, with a total of six plots irrigated with TWW and six plots irrigated with FW. The Eucalyptus trees in sites 3 and 4 were irrigated using the immersion method, which is common for perennial plants in El-Oued City.

The Sorghum plots in sites 1 and 2 were irrigated with a drip irrigation system. Each plot of Sorghum covered an area of 12 m^2 (3 m × 4 m). The Sorghum seeding was rated at 25 seeds/m^2 without using any tillage technique. The row distance was 75 cm and spacing was 8 cm between the seeds to prevent weed growth. The first sowing took place on the 1st of September 2018 due to the appropriate temperatures in the experimental region. In addition, the experimental sites have not been previously planted or composted. Moreover, the Sorghum experiments were conducted over 04 campaigns in two different seasons: Autumn and Spring. The first campaign in 2018 occurred from September to November. The second campaigns were in March–May 2019 and September–November 2019. The last campaign took place in March 2020. The Sorghum sampling was performed randomly during the four campaigns. The fresh Sorghum sampled parts were collected in the maturing stage of each campaign and then carried to the laboratory for analysis.

The Eucalyptus plantation irrigated with TWW (site 3) had an area of 1.06 ha, whereas the Eucalyptus plantation irrigated with FW (site 4) covered 0.65 ha. Every tree site was divided into 03 repetitions plots. The sampling was conducted from 2018 to 2020. The tree sampling was done using a simple random method in consideration of the age factor. The selected ages were 3, 7, and 10 years old in every site. The number of the sampled trees was 54. The sampling of leaves and branches was carried out by classifying each sampled tree into three height layers: 0–1 m, 1–2 m, and 2–4 m. The Eucalyptus leaves were randomly sampled from the trees in each site. The branches were sized on the trees depending on the studied layers. The diameter measures were taken by a caliper. The number of measured branches per tree varies by layer, ranging from 6 to 55; all the branches in each layer were included in the measurement. The total number of the measured branches was 1339. All branch samples were measured in each of the sampled trees at one time, and all sizings were completed within 15 days. Furthermore, the soil of the Eucalyptus sites had been irrigated for 12 years when the experiment was established in 2018.

Irrigation schedule

The irrigation duration for Eucalyptus was 1 h per day twice a week in autumn and winter. In spring, the duration was 1 h three times a week, while in summer, it was 1 h per day. The flow rate was 90 L/min for all irrigated plots. In the Sorghum sites, the watering frequency was three to four times a week for 1 h per day in autumn and winter. At spring, the frequency was 1 h every 2 days. In summer, watering occurred every day for 4 h. The combined flow rate for every plot was 20 L per 30 min.

Water sampling, conservation, and analysis

Water sampling was carried out during autumn and spring in three experimental years (2018, 2019, and 2020). The Water samples were collected during the irrigation process. For every water type, six repetitions were conducted for each year. The storage of water samples was in 500 mL plastic PET bottles at a temperature of 4 °C. After that, these samples were transported to the quality control and fraud prevention laboratory in El-Oued. At this laboratory, the samples were prepared by filtration using Whatman paper to 0.2 µm, 10 µm, and 16 µm.

The TWW samples were analyzed in different laboratories. The laboratory of the Kouinine WWTP (Algeria) determined N as reported by the standard method of Rodier et al. (2009). Electrical conductivity (EC) and pH were determined there as well using a Tetracon pH 510-m and a conductivity meter Terminal 740. FW analysis was performed in the municipal water control laboratory in El-Oued. pH and EC were analyzed using the NF T90-008 and NF T90-031 methods, respectively. The NO_3 was measured using the sodium salicylate method.

TWW samples were acidified to pH 1–2 with HNO_3 (65%), then filtered in compliance with the DIN EN ISO 11,885/DEV E22 standard at the UFZ Helmholtz Centre for Environmental Research GmbH's analytical chemistry laboratory in Germany. After that, the concentration of calcium (Ca), magnesium (Mg), and sodium (Na) were assayed. At the Agronomy Department laboratory at the Faculty of Agricultural and Environmental Sciences, Rostock University, Germany, potassium (K) and phosphorus (P) values were analyzed according to DIN EN ISO 11,885 with an ICP-MS8800 system.

Estimation of Eucalyptus biomass

The calculation of Eucalyptus branch biomass (BB) was based on the formula of[64] as follows:

$$BBn = EXP^{(-2.61184+2.34891 lnBDn)} \tag{1}$$

where BBn is the number of the individual branch, and BDn is the single branch diameter.

The biomass values of the single branches in a certain height layer and tree age were then summed up to get the branch biomass (BB) for the section described. 03 replicates for every tree age were applied.

The evaluation of the Eucalyptus aboveground biomass (EAB) was performed by measuring the diameter at breast height and the total tree height using a measuring tape, forestry diameter tape, and a Blume–Leiss Dendrometer. The EAB of the sampled trees was calculated using the allometric growth Equation[65]:

$$EAB = 0.02407389(D^2H)^{0.9768058} + 0.00492556(D^2H)^{0.8449044} + 0.0007088(D^2H)^{0.935545} + 0.0063525(D^2H)^{0.8738162} \tag{2}$$

where D indicates the diameter at breast height (cm), and H is the total Eucalyptus tree height (m).

The carbon content in the sampled branches was estimated by calculating the mean values of this parameter from various research data[66–68]. Formula (3) was used to calculate the total biomass carbon of the Eucalyptus branches (TBCB) in each height layer and age category in all the sampled trees, accordingly:

$$TBCB = (Cb * BBn) \tag{3}$$

where Cb = 48.70%, i.e., the mean total carbon content of Eucalyptus woody parts.

Calculation of Sorghum biomass

For Sorghum measurements, 03 replicates were conducted for each campaign. The Sorghum aboveground biomass (SAB) was estimated via its aboveground DM. The biomass analysis was conducted in the Laboratory of Quality Control and Fraud Prevention based on drying techniques. The samples were dried in open air for 24 h, then dried at 45 °C for 48 h in the oven. Subsequently, the dried aerial parts of Sorghum were weighed. The calculation of the DM was performed using this formula (4):

$$SAB = MF/MS * 100 \tag{4}$$

where MF is the fresh weight, and MS is the dry weight of the Sorghum plant.

The total carbon biomass of Sorghum (STCB) was calculated after a chemical analysis of the total carbon (CTot) content. The following Eq. (5) was used:

$$STCB = C_{Tot} * SAB \tag{5}$$

where the total carbon (C_{Tot}) analysis was performed using the infrared spectroscopy method with an ELTRA CS-580A analyzer.

Nitrogen analysis

To check the nitrogen content, samples of the aboveground Sorghum and Eucalyptus were weighed fresh, and dried using an analytical balance KERN ACJ/AES. The oven-drying process was performed at 55 °C for 24 h. After that, the dried samples were milled and conserved in Kraft bags. The analyses of the samples were performed at the Agricultural Chemistry and Environmental Biogeochemistry Laboratory at Poznan University of Life Sciences in Poland. The total nitrogen (N_{Tot}) was determined according to the titration procedure using the Kjeldahl method.

Type of water	Descriptive statistics	Ca (mg/L)	Mg	Na	K	N–NO_3	P	pH	EC* (mS/cm)
Freshwater (FW)	Min	0.32	0.11	0.51	40.26	0.67	0.13	7.15	2.47
	Max	0.73	0.16	0.83	44.08	0.98	0.26	7.70	4.15
	Mean	**0.47**	**0.12**	**0.69**	**42.05**	**0.82**	**0.19**	**7.43**	**3.44**
	SD[a]	**0.18**	**0.01**	**0.12**	**0.89**	**0.10**	**0.03**	**0.14**	**0.42**
	CV[b]	38.7	9.9	0.55	2.1	12.6	15.4	1.9	12.3
Treated wastewater (TWW)	Min	0.36	0.12	0.55	82.78	0.97	1.31	6.67	4.97
	Max	0.43	0.15	0.67	89.07	3.97	2.41	8.03	6.18
	Mean	**0.38**	**0.13**	**0.56**	**86.93**	**1.96**	**1.70**	**7.33**	**5.41**
	SD	**0.02**	**0.01**	**0.02**	**1.42**	**0.89**	**0.35**	**0.42**	**0.21**
	CV	5.1	3.6	3.4	1.6	45.4	20.5	5.7	3.9

Table 1. Selected chemical parameters of water used in the study. [a]Standard deviation (n = 54); [b]Coefficient of variation; *Electrical conductivity Significance values are in bold.

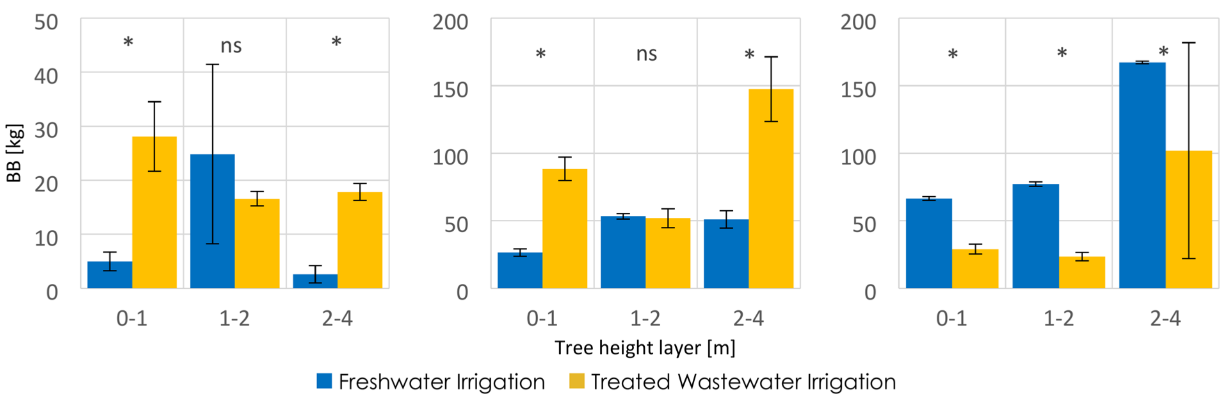

Fig. 2. Eucalyptus branch biomass (BB) in different height layers at different tree ages. Tree ages from left to right: 3 years old, 7 years old, and 10 years old. Vertical bars show standard deviation (SD). Symbols above each compared group indicate significance (ns—no significant difference, * $p \le 0.05$).

Statistical analysis

Firstly, Microsoft Excel (Version 2411 Build 16.0.18227.20082) was exploited to manage, sort, and calculate the data. Levene's test and Shapiro–Wilk test were performed to check the homogeneity and the normality of the variance for each group of dependant variables. The software SPSS version 22 was used for the ANOVA tests, Tukey tests, independent samples t-test, and Mann–Whitney U test. In addition, Kruskal-Wallis tests, Dunn's tests with a Holm-Bonferroni adjustment method were performed using the R language (version 4.4.2) with the FSA package. The biomass data of the Eucalyptus branches were not normally distributed. They were analyzed using the Kruskal–Wallis test followed by the Mann–Whitney U post hoc test. The results were interpreted based on standard significance thresholds ($p \le 0.05$).

Results

Water characteristic

Compared to FW, the TWW had a similar Mg content, double the K content, and a P content nearly nine times as high. The mean values of these elements content in FW and TWW were 0.12 mg/L and 0.13 mg/L for Mg, 42.05 mg/L and 86.93 mg/L for K, and 0.19 mg/L and 1.70 mg/L for P, respectively (Table 1). For Ca, the mean values were lower in the TWW in comparison to the FW with contents of 0.38 mg/L and 0.47 mg/L, respectively. The nitrate-nitrogen (N–NO_3) was higher in TWW (1.96 mg/L) than in FW(0.82 mg/L). The pH values did not show a marked difference, but EC values were higher in the TWW with a mean of 5.41 mS/cm, while EC had a mean of 3.44 mS/cm in FW. Only a minimal Na concentration of 0.02 mg/L was found in the TWW, whereas it was 0.69 mg/L in the FW.

Dynamics of branch biomass in Eucalyptus

The Kruskal–Wallis test proved that one or more of the three independent factors (water type, tree age, tree height) significantly affected ($p = 0.00006$) BB dynamics.

Figure 2 demonstrated that FW positively impacted BB growth at heights 1–2 m at 3 and 7 years old trees. However, these results showed no statistically significant difference compared to the impact of TWW irrigation. Nevertheless, there was a significant difference in BB growth at heights of 0–1 m. For 3 years old trees, mean values were 28.10 kg for TWW and 4.98 kg for FW. For 7 years old trees, mean values were 88.38 kg and 26.42 kg for TWW and FW, respectively. Similarly, results showed significantly ($p \leq 0.05$) higher BB mean values for TWW compared to FW irrigation at heights 2–4 m, with mean values of 17.83 kg and 2.58 kg for the 3 years old trees, and 147.39 kg and 50.93 kg for the 7 years old trees, respectively.

In contrast, for 10 years old trees, the mean values of BB were significantly higher under FW than TWW irrigation across all tree heights. At a height of 0–1 m, BB mean values were 66.45 kg and 28.97 kg. And they were 77.14 kg and 23.46 kg at a height of 1–2 m. These values were 167.19 kg and 101.89 kg at a height of 2–4 m, for FW and TWW, respectively.

Aboveground biomass distribution in Eucalyptus

The Eucalyptus aboveground biomass measurements followed a normal distribution, as stated by the Shapiro–Wilk test. The two-way ANOVA test indicated significant differences ($p = 0.000$) in EAB across different tree ages and irrigation types. Even though the 3-year-old Eucalyptus trees showed higher EAB when irrigated with TWW (22.79 kg) in comparison with FW (6.56 kg) irrigation (Fig. 3), this difference was not statistically significant according to the Tukey test. However, the opposite applied to 7 and 10 years old trees. For the 7 years old trees, the EAB mean values ranged between 126.64 kg for TWW irrigation and 204.56 kg for FW irrigation. These results were confirmed to be statistically significant ($p \leq 0.05$). Additionally, for the 10 years old trees, there was a highly significant difference ($p \leq 0.001$) in EAB under FW irrigation (Fig. 3). The mean values were 222.56 kg and 364.30 kg for TWW and FW irrigation, respectively.

Dynamics of total branch carbon biomass in Eucalyptus

The results presented the effectiveness of water types in modifying TBCB dynamics (Fig. 4). The irrigation with TWW increased the TBCB at height layers 0–1 m in the 3 years old and 7 years old trees. Similarly, TBCB was higher at height layers 2–4 m at the same trees ages. At the age of 3 years old, the mean values were 13.68 kg under TWW irrigation and 2.42 kg under FW irrigation at the height of 0–1 m, whereas they were 8.68 kg for TWW and 1.25 kg for FW at 2–4 m tree height. In the case of trees aged 7 years, TWW irrigation gave a higher mean value (43.04 kg) compared to the FW irrigation (12.86 kg) at the height of 0–1 m. Likewise, the mean values were higher in the TWW irrigation (71.78 kg) than in the FW irrigation (24.80 kg) at 2–4 m tree height. At height layer 1–2 m, the mean values were quite similar for both irrigation types. For the 10 years old trees, Fig. 4 showed a decrease in TBCB production under TWW irrigation. The mean values were approximately half what they were under FW irrigation.

Dynamics of nitrogen in Eucalyptus leaves

The Kruskal–Wallis test revealed a statistically significant difference ($p = 0.00002$) in the total nitrogen content of Eucalyptus leaves depending on the categorial variables: water types, tree ages, and tree heights. The Mann–Whitney U tests confirmed these significant differences ($p \leq 0.05$) between FW and TWW irrigation types.

Results (Fig. 4) showed that N_{Tot} in Eucalyptus leaves was significantly higher ($p \leq 0.05$) in all trees irrigated with TWW compared to those irrigated with FW.

At 3 years old trees, the N_{Tot} mean values were 2.56% and 1.44% at a height of 0–1 m, and 1.94% and 1.33% at a height of 1–2 m. Also, the values were 2.04% and 1.48% at a height of 2–4 m for TWW and FW irrigation, respectively. Likeness, at 7 years old trees, and a height of 0–1 m, the mean values of N_{Tot} in Eucalyptus leaves were 2.62% for TWW and 2.07% for FW irrigation. At a height of 1–2 m, they were 2.84% and 1.82% for TWW and FW irrigation, respectively. Moreover, the mean values were 3.10% for TWW irrigation and 2.33% for FW irrigation at a height of 2–4 m.

In addition, the 10 years old trees indicated mean N_{Tot} values of 2.08% under TWW irrigation and 1.82% under FW irrigation at a layer height of 0–1 m. At layer height 1–2 m, the values were 2.18% for TWW irrigation and 1.74% for FW irrigation. Lastly, at a layer height of 2–4 m, the mean values were 1.65% and 2.36% for FW and TWW irrigation, respectively.

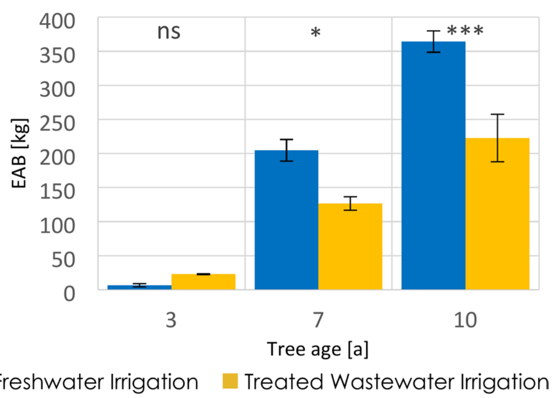

Fig. 3. Eucalyptus aboveground biomass (EAB) dynamics and distribution at different tree ages. Vertical bars show standard deviation (SD). Symbols above each compared group indicate significance (ns—no significant difference, * $p \leq 0.05$, *** $p \leq 0.001$).

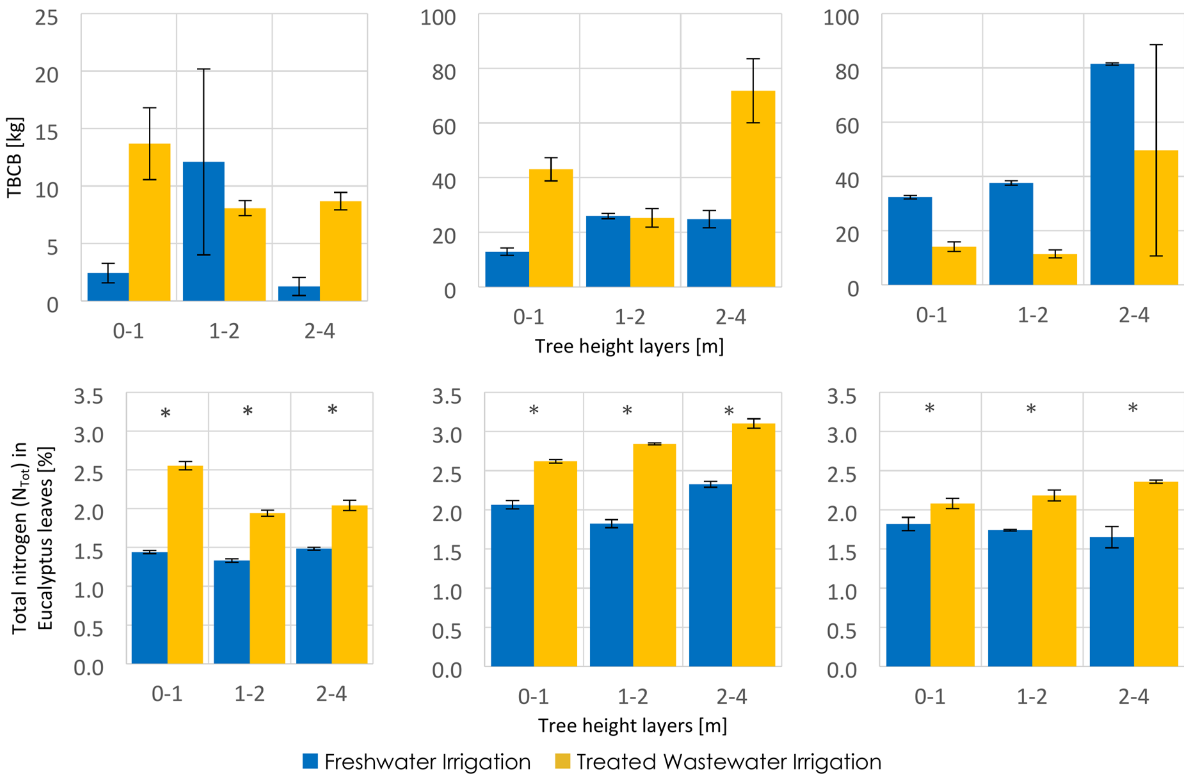

Fig. 4. Total biomass carbon of Eucalyptus branches (TBCB, top) and total nitrogen (N_{Tot}) concentration in Eucalyptus leaves (bottom) in different layers at different tree ages. Tree ages from left to right: 3 years old, 7 years old, and 10 years old. Vertical bars show standard deviation (SD). Symbols above each compared group indicate significance (* $p \leq 0.05$).

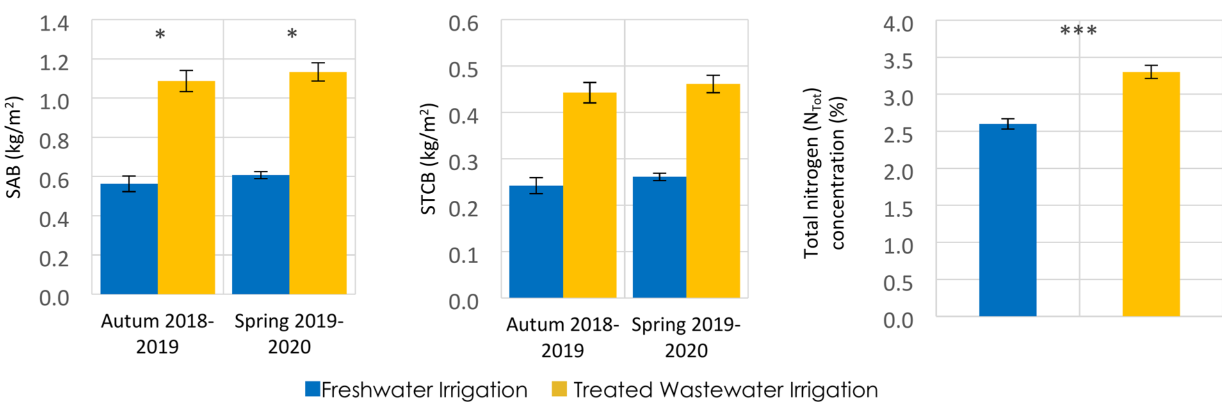

Fig. 5. Sorghum aboveground biomass (SAB) in different growing seasons (left). Total carbon biomass in Sorghum (STCB) in different growing seasons (middle), and total nitrogen (N_{Tot}) concentration (right) in Sorghum aboveground biomass (SAB). Vertical bars are standard deviations. Symbols above each compared group indicate significance (* $p \leq 0.05$, *** $p \leq 0.001$).

Consequently, TWW irrigation positively impacted the rise of N_{Tot} content in Eucalyptus leaves at the three studied tree heights and ages.

Biomass dynamics and distribution in Sorghum

The Sorghum aboveground biomass was significantly ($p \leq 0.05$) higher in the TWW irrigation site compared to the FW irrigation site (Fig. 5). During autumn, SAB values were significantly higher by 1.08 kg/m^2 in the TWW irrigated sites compared to FW irrigated sites by 0.56 kg/m^2. In spring, the SAB values remained significantly higher in TWW irrigated sites when compared with FW irrigated sites (1.13 kg/m^2 and 0.60 kg/m^2, respectively.).

The results of carbon biomass distribution revealed that TWW irrigation substantially increased the STCB content in all the studied seasons, the mean values were nearly double compared to the STCB in the FW irrigation sites. In autumn, the mean values varied within 0.24 kg/m^2 and 0.44 kg/m^2 for the FW and the TWW irrigated sites, respectively. Similar results were shown in spring, where TWW irrigation developed better the STCB (0.46 kg/m^2) than the FW irrigation (0.26 kg/m^2).

For the Sorghum N_{Tot} data, the values were confirmed to follow a normal distribution. The one-way ANOVA followed by a t-test gave a highly significant difference ($p \leq 0.001$) was noted in N_{Tot} concentration between the Sorghum irrigated with distinct water types (Fig. 5). The irrigation with TWW affected significantly (t-test $p \leq 0.001$) the concentration of N_{Tot}. As a consequence, this water type boosted the N_{Tot} in Sorghum to 3.30%, compared to 2.60% in Sorghum irrigated with FW.

Discussion

The results supported our first hypothesis and concurred with those reported in existing research[41]. The study confirmed a progressive development of Eucalyptus branch biomass growth through TWW irrigation and in early tree ages. This growth was seen over the Eucalyptus branch biomass development in the studied heights of 3 and 7 years old, while the 10 years old Eucalyptus didn't exhibit this trend. The significant results ($p \leq 0.05$) showed a lower response in Eucalyptus branch biomass production for the case of 10 years old trees at all layer heights under TWW irrigation. According to previous results[69], TWW contains various nutrients that can improve Eucalyptus branch biomass growth. This water type plays a fertilizer role by providing the major minerals like N, P, and K.

Also, the study outcomes indicated that the lower EC concentrations in FW (mean EC = 3.44 mS/cm) may lead to significantly ($p \leq 0.05$, and $p \leq 0.001$) better EAB growth in older Eucalyptus trees compared to the ones irrigated with TWW (mean EC = 5.41 mS/cm), which had higher EC concentrations.

Subsequently, the total biomass carbon of the Eucalyptus branches results reflected the amount of carbon sequestrated in different height layers and tree ages under TWW and FW irrigation. Carbon sequestration occurs by assimilating atmospheric CO_2 and transforming it into carbohydrates[70]. According to the results, TWW improved the biomass growth in Eucalyptus branches more than FW, especially in the Eucalyptus young trees at 0–1 m and 2–4 m. parallel to[71,72] researchs, TWW irrigation could also contribute to mitigating climate change by carbon branch storage.

In addition, our findings revealed that the concentration of N_{Tot} in Eucalyptus was significantly higher ($p \leq 0.05$) in all plots irrigated with TWW compared to those irrigated with FW. This macronutrient was better stored in Eucalyptus leaves at all heights and ages when TWW was applied, which may be explained by the higher $N\text{–}NO_3$ content in this recycled water compared to FW. In line with existing statements[27], our study confirmed that TWW irrigation increased the nitrogen content in the studied trees and may boost biomass production in Eucalyptus leaves.

For Sorghum, TWW irrigation considerably enhanced ($p \leq 0.05$) the growth dynamics of the Sorghum aboveground biomass and gave significantly higher yields than in the case of FW irrigation. Previous work[73] confirmed that TWW contributed 21% of the crops' potassium requirements, while FW contributed only 7%. In line with this research, the study results showed a higher potassium content in TWW (mean $K^+ = 86.93$ mg/L) compared to FW potassium with a mean $K^+ = 42.05$ mg/L. This may explain the higher increase of the Sorghum aboveground biomass due to the K^+ nutrient provided by TWW irrigation in the two growing seasons. Furthermore, the results showed that TWW had a significant amount of phosphorus (mean P = 1.70 mg/L). In contrast, FW had a lower P content (mean P = 0.19 mg/L). This mineral is indispensable for plant growth[22]. The higher P concentrations found in TWW could produce better Sorghum aboveground biomass and improve Sorghum crop performance more than FW. A sequel to that, the results showed that the Sorghum aboveground biomass could be affected positively by weather conditions[74], especially in arid environments. The higher growth in the Sorghum aboveground biomass in spring season can be explained by the rise in temperature, which became more suitable for this plant than autumn temperatures. Accordingly, the study also demonstrated that Sorghum adapts well to the climatic changes in the Saharan conditions. The results confirmed the findings of previous research[22], as TWW irrigation was found to stimulate Sorghum plants to accumulate high concentrations of macro and microelements, thus building up more biomass and storing green energy in the arid areas. Khaskhoussy et al.[69] have confirmed that the aboveground biomass can be increased due to calcium and magnesium elements transferred from TWW to the irrigated plants and contribute to better Sorghum growth.

Regarding the total carbon biomass of Sorghum, total carbon results under TWW irrigation were the highest during all studied seasons. The carbon content in trees could be impacted by climatic factors[75]. It appears that the warm weather may contribute to the notable growth of the total carbon biomass of Sorghum under TWW irrigation, where Saharan temperatures in autumn and spring favor this amelioration[76].

The significant ($p \leq 0.001$) high N concentration in Sorghum under TWW irrigation contributed greatly to the development of Sorghum biomass as a major element in its productivity growth[77].

Our study emphasized N and C dynamics in the investigated plants due to their big role in the production of DM. They present the most important elements that can form biomass[36]. The quantitative estimation of carbon stocks in Eucalyptus branches is crucial as it serves to trace potential carbon assimilation by the tree under degraded lands in hyper-arid regions. Based on previous research[78] and through the irrigation process, our results affirm that the fertilizing property of TWW can also stimulate total carbon build-up in Eucalyptus branches, and nitrogen accumulation in Eucalyptus leaves at different heights and young ages.

The limitations of the Eucalyptus branch biomass and Eucalyptus aboveground biomass estimation models in this research may arise from the different tree architectures and shapes, as well as the different genus and species of the studied Eucalyptus trees[64]. Also the low efficiency of the Blume–Leiss method for assessing tree height [79].

Conclusions

In the desert regions of the country, Algerian authorities are focusing on reusing TWW to address the shortage of FW. This paper can help provide insight by examining the effect of using TWW for irrigation on the biomass and nitrogen dynamics of Eucalyptus and Sorghum. The results showed a positive influence of TWW on biomass growth, although this is more evident in Sorghum than in Eucalyptus. The same applies to the C storage when comparing the two types of irrigation. A clearly positive influence of TWW irrigation compared to FW irrigation is particularly obvious in the N content.

The detailed investigation in different height layers and across different tree ages of Eucalyptus provided an insight into the dynamics of biomass, carbon, and N contents in these trees. Even though an increase in biomass due to TWW irrigation was clearly visible in Eucalyptus, an increased concentration of C and N was found. There is promising potential to meet the high water demand of Eucalyptus plantations and conserve FW resources through TWW reuse.

Sorghum biomass growth and increase in C and N concentration demonstrated a remarkably positive response under TWW irrigation, probably influenced by the raised nutrient availability. This outcome could be observed over several cultivation periods. TWW saves valuable FW resources, allowing the studied plants to be grown for windbreaking, biomass production and carbon sequestration. This can contribute to combating desertification, especially in hyper-arid regions.

Data availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Received: 25 August 2024; Accepted: 29 April 2025
Published online: 27 May 2025

References

1. Heggy, E., Bermudez, V. & Vermeersch, M. *Sustainable Energy- Water-Environment Nexus in Deserts*. https://doi.org/10.1007/978-3-030-76081-6 (2021).
2. Tzanakakis, V. A., Paranychianakis, N. V. & Angelakis, A. N. Water supply and water scarcity. *Water (Switzerland)* **12**, 1–16 (2020).
3. Tyczewska, A., Woźniak, E., Gracz, J., Kuczyński, J. & Twardowski, T. Towards food security: Current state and future prospects of agrobiotechnology. *Trends Biotechnol.* https://doi.org/10.1016/j.tibtech.2018.07.008 (2018).
4. Rizzo, L. et al. Best available technologies and treatment trains to address current challenges in urban wastewater reuse for irrigation of crops in EU countries. *Sci. Total Environ.* **710**, 136312 (2020).
5. Liu, W., Liu, X., Yang, H., Ciais, P. & Wada, Y. Global water scarcity assessment incorporating green water in crop production. *Water Resour. Res.* https://doi.org/10.1029/2020WR028570 (2022).
6. Hamiche, A. M., Stambouli, A. B. & Flazi, S. A review on the water and energy sectors in Algeria: Current forecasts, scenario and sustainability issues. *Renew. Sustain. Energy Rev.* **41**, 261–276 (2015).
7. Sekkoum, K. et al. Water in Algerian Sahara: Environmental and health impact. *Adv. Desalin.* https://doi.org/10.5772/50319 (2012).
8. Tigrine, Z. et al. Membrane desalination technology in Algeria: Reverse osmosis for coastal areas. *Green Energy Technol.* https://doi.org/10.1007/978-3-319-62575-1_15 (2018).
9. Viala, E. Making the most of scarcity accountability for better water management in the Middle East and North Africa: Julia Bucknall et al., 2007, World Bank, ISBN-13: 978-0-8213-6925-8. *Irrig. Drain. Syst.* **22**(2), 189–191. https://doi.org/10.1007/s10795-008-9048-4 (2008).
10. Touitou, M. & Al-Amin, A. Q. Climate change and water resources in Algeria: Vulnerability, impact and adaptation strategy. *Econ. Environ. Stud.* **18**, 411–429 (2018).
11. Nicolas R., D. & Saeid, E. Environmental impacts of drought on desertification classification. in *Handbook of Drought Water Scarcity* 45–64 https://doi.org/10.1201/9781315226781-3 (2017).
12. Singh, K. Managing water for sustainable development: An economist's perspective. *IIM Kozhikode Soc. Manag. Rev.* https://doi.org/10.1177/2277975215619564 (2016).
13. Gude, V. G. Desalination and water reuse to address global water scarcity. *Rev. Environ. Sci. Biotechnol.* **16**, 591–609 (2017).
14. Sileshi, G. W., Gupta, S. R. & Dagar, J. C. *Synthesis: Agroforestry for Sustainable Intensification of Agriculture in Asia and Africa*. (2023). https://doi.org/10.1007/978-981-19-4602-8_25
15. Alherbawi, M., McKay, G., Govindan, R., Haji, M. & Al-Ansari, T. A novel approach on the delineation of a multipurpose energy-greenbelt to produce biofuel and combat desertification in arid regions. *J. Environ. Manag.* **323**, 116223 (2022).
16. Oulmane, A., Kechar, A., Benmihoub, A. & Amine Benmehaia, M. Assessment of surface water management institutions: A case of public irrigation schemes in northern Algeria. *Water Policy* **00**, 1–13 (2022).
17. Dairi, S., Mrad, D., Boukhari, S., Djebbar, Y. & Habib, A. Prospective scenarios for the management of wastewater resources in mostaganem region Algeria. *Desalin. Water Treat.* **209**, 414–428 (2021).
18. Bouklia, M.H. *Algérie, Rapport National Volontaire 2019. Progression de la mise en oeuvre des ODD.* (2019).
19. Djemil, W., Hannouche, M. & Belksier, M. S. Reuse of treated wastewater in agriculture: Physicochemical quality and environmental risks. Case of wastewater treatment plant of Baraki and Beni Messous. Algeria. *AIP Conf. Proc.* **1968**, (2018).
20. Boudries, N. et al. Physicochemical and functional properties of starches from sorghum cultivated in the Sahara of Algeria. *Carbohydr. Polym.* **78**, 475–480 (2009).
21. Kimera, F., Mugwanya, M., Madkour, K., Dawood, M. A. O. & Sewilam, H. Maximization of brackish water productivity for the sustainable production of striped catfish (*Pangasianodon hypophthalmus*) and grain sorghum (*Sorghum bicolor* (L.) Moench) cultivated under an integrated aquaculture–agriculture system. *Environ. Sci. Pollut. Res.* **31**, 31878–31895 (2024).
22. Seleiman, M. F. et al. Impacts of long-and short-term of irrigation with treated wastewater and synthetic fertilizers on the growth, biomass, heavy metal content, and energy traits of three potential bioenergy crops in arid regions. *Energies* **14**, 1–22 (2021).
23. Habyarimana, E., Laureti, D., De Ninno, M. & Lorenzoni, C. Performances of biomass sorghum [*Sorghum bicolor* (L.) Moench] under different water regimes in Mediterranean region. *Ind. Crops Prod.* **20**, 23–28 (2004).
24. Taia, A. et al. Consecutive seasonal effect on yield and water productivity of drip deficit irrigated sorghum in saline soils. *Saudi J. Biol. Sci.* **29**, 2683–2690 (2022).
25. Bakari, H. et al. Sorghum (*Sorghum bicolor* L. Moench) and its main parts (by-products) as promising sustainable sources of value-added ingredients. *Waste Biomass Valoriz.* **14**, 1023–1044 (2023).

26. Ganjegunte, G., Ulery, A., Niu, G. & Wu, Y. Treated urban wastewater irrigation effects on bioenergy sorghum biomass, quality, and soil salinity in an arid environment. *L. Degrad. Dev.* **29**, 534–542 (2018).
27. Maucieri, C. et al. Sorghum biomass production for energy purpose using treated urban wastewater and different fertilization in a mediterranean environment. *Agriculture* **6**, 1–15 (2016).
28. Quan, C., Zhou, Y., Wang, J., Wu, C. & Gao, N. Biomass-based carbon materials for CO2capture: A review. *J. CO2 Util.* **68**, 102373 (2023).
29. Wang, Y. et al. Biomass-derived carbon materials: Controllable preparation and versatile applications. *Small* **17**, 1–32 (2021).
30. Behera, L., Ray, L. I., Ranjan Nayak, M. & Mehta, A. Carbon sequestration potential of Eucalyptus spp.: A review. *E-Planet* **18**, 79–84 (2020).
31. Kumar, P. et al. Carbon sequestration and soil carbon build-up under eucalyptus plantation in semi-arid regions of North-West India. *J. Sustain. For.* **40**, 319–331 (2021).
32. Ning, P. et al. Recent advances in the valorization of plant biomass. *Biotechnol. Biofuels* **14**, 1–22 (2021).
33. Rademacher, P., Khanna, P. K., Eichhorn, J. & Guericke, M. Tree growth, biomass, and elements in tree components of three beech sites. In *Functioning and Management of European Beech Ecosystems* (eds Brumme, R. & Khanna, P. K.) 105–136 (Springer Berlin Heidelberg, 2009). https://doi.org/10.1007/b82392_9.
34. Singh, J., Singh, B. & Sharma, S. Comparison of soil carbon and nitrogen pools among poplar and eucalyptus based agroforestry systems in Punjab India. *Carbon Manag.* **12**, 693–708 (2021).
35. Soltani-Gerdefaramarzi, S., Alemzadeh, A., Yarami, N. & Dehestani-Ardakani, M. The effects of irrigation methods and water quality on soil chemical properties and leaf nutrient uptake of lavandula (*Lavandula angustifolia* L.) in an arid region. *J. Plant Growth Regul.* **42**, 1256–1265 (2023).
36. Sharrow, S. H. & Ismail, S. Carbon and nitrogen storage in agroforests, tree plantations, and pastures in western Oregon USA. *Agrofor. Syst.* **60**, 123–130 (2004).
37. SZekri, S. *Water Policies in MENA Countries* (Springer International Publishing, 2020).
38. Saadaoui, E., Ben Yahia, K., Dhahri, S., Ben Jamaa, M. L. & Khouja, M. L. An overview of adaptative responses to drought stress in Eucalyptus spp.. *For. Stud.* **67**, 86–96 (2017).
39. Zhang, H., Jiang, Y., Song, M., He, J. & Guan, D. Improving understanding of carbon stock characteristics of Eucalyptus and Acacia trees in southern China through litter layer and woody debris. *Sci. Rep.* **10**, 1–11 (2020).
40. Cherfouh, R., Yves, L., Derridj, A. & Merdy, P. Long-term, low technicality sewage sludge amendment and irrigation with treated wastewater under Mediterranean climate: Impact on agronomical soil quality. *Environ. Sci. Pollut. Res.* **25**, 35571–35581 (2018).
41. Maaloul, A. et al. Effect of treated wastewater on growth and secondary metabolites production of two Eucalyptus species. *Agric. Water Manag.* **211**, 1–9 (2019).
42. FAO. *Les eucalyptus dans les reboisements. L'Organisation des Nations unies pour l'alimentation et l'agriculture* **11**, (1982).
43. Boulmane, M., Oubrahim, H., Halim, M., Bakker, M. R. & Augusto, L. The potential of Eucalyptus plantations to restore degraded soils in semi-arid Morocco (NW Africa). *Ann. For. Sci.* https://doi.org/10.1007/s13595-017-0652-z (2017).
44. Jaleta, D., Mbilinyi, B., Mahoo, H. & Lemenih, M. Eucalyptus expansion as relieving and provocative tree in Ethiopia. *J. Agric. Ecol. Res. Int.* **6**, 1–12 (2016).
45. Mosa, A., Hawamdeh, O. A., Rady, M. & Taha, A. A. Ecotoxicological monitoring of potentially toxic elements contamination in Eucalyptus forest plantation subjected to long-term irrigation with recycled wastewater. *Environ. Pollut.* **329**, 121739 (2023).
46. Chavan, S. B. et al. Carbon sequestration potential of commercial agroforestry systems in indo-gangetic plains of India: Poplar and Eucalyptus-based agroforestry systems. *Forests* **14**, 1–23 (2023).
47. Chemlali, I. et al. Anatomical, physiological, biochemical and molecular responses of Eucalyptus spp. under water deficit conditions and characteristics of Tunisian arid species: An overview. *Not. Sci. Biol.* **14**, 1–20 (2022).
48. Pedrero, F., Kalavrouziotis, I., Alarcón, J. J., Koukoulakis, P. & Asano, T. Use of treated municipal wastewater in irrigated agriculture-review of some practices in Spain and Greece. *Agric. Water Manag.* **97**, 1233–1241 (2010).
49. Al-Suhaibani, N. et al. Integrative effects of treated wastewater and synthetic fertilizers on productivity, energy characteristics, and elements uptake of potential energy crops in an arid agro-ecosystem. *Agronomy* **11**, 2250 (2021).
50. Huang, J. & Hartemink, A. E. Soil and environmental issues in sandy soils. *Earth-Sci. Rev.* **208**, 103295 (2020).
51. Elbana, T. A., Bakr, N. & Elbana, M. Reuse of treated wastewater in Egypt: Challenges and opportunities. *Handb. Environ. Chem.* **75**, 429–453 (2019).
52. Mainardis, M. et al. Wastewater fertigation in agriculture: Issues and opportunities for improved water management and circular economy. *Environ. Pollut.* **296**, 2022 (2022).
53. Erel, R. et al. Long-term irrigation with reclaimed wastewater: Implications on nutrient management, soil chemistry and olive (*Olea europaea* L.) performance. *Agric. Water Manag.* **213**, 324–335 (2019).
54. Singh, A. A review of wastewater irrigation: Environmental implications. *Resour. Conserv. Recycl.* **168**, 105454. https://doi.org/10.1016/j.resconrec.2021.105454 (2021).
55. Bardhan, G., Russo, D., Goldstein, D. & Levy, G. J. Changes in the hydraulic properties of a clay soil under long-term irrigation with treated wastewater. *Geoderma* **264**, 1–9 (2016).
56. Bedbabis, S., Ferrara, G., Ben Rouina, B. & Boukhris, M. Effects of irrigation with treated wastewater on olive tree growth, yield and leaf mineral elements at short term. *Sci. Hortic. (Amsterdam)* **126**, 345–350 (2010).
57. Helmecke, M., Fries, E. & Schulte, C. Regulating water reuse for agricultural irrigation: Risks related to organic micro-contaminants. *Environ. Sci. Eur.* https://doi.org/10.1186/s12302-019-0283-0 (2020).
58. Khezzani, B. & Bouchemal, S. Development and conservation of water resources for agriculture in an arid environment: A case study of the souf oasis (Algerian Sahara). *Ann. Arid Zone* **57**, 1–11 (2018).
59. Meftah, N. & Mahboub, M. S. Spectroscopic characterizations of sand dunes minerals of El-Oued (Northeast Algerian Sahara) by FTIR XRF and XRD analyses. *SILICON* **12**, 147–153 (2020).
60. Remini, B. le Souf: Quand le forage et le pivot menacent le ghout!. *Larhyss J.* **37**, 23–38 (2019).
61. Zeghdi, S., Bebba Ahmed, A. & Laouini, S. E. Chemistry wastewater treatment of El-Oued City (South-east of Algeria) by utilization of Typha latifolia. *Orient. J. Chem.* **32**, 3043–3052 (2016).
62. Bedbabis, S., Ben Rouina, B., Boukhris, M. & Ferrara, G. Effect of irrigation with treated wastewater on soil chemical properties and infiltration rate. *J. Environ. Manag.* **133**, 45–50 (2014).
63. Soumaia.H et Hakima.K. Etude de D iversité floristique d u chott Edhiba (région du Souf , Sahara septentrional Est). (Université Echahid Hamma Lakhdar -El OUED, 2019).
64. Garcia Florez, L., Vanclay, J. K., Kevin Glencross, J. & Nichols, D. Developing biomass estimation models for above-ground compartments in Eucalyptus dunnii and Corymbia citriodora plantations. *Biomass Bioenergy* **130**, 105353. https://doi.org/10.1016/j.biombioe.2019.105353 (2019).
65. Liu, Y. et al. Individual tree aboveground biomass estimation based on UAV stereo images in a Eucalyptus plantation. *Forests* **14**, 1–22 (2023).
66. Paul, K. I., Jacobsen, K., Koul, V., Leppert, P. & Smith, J. Predicting growth and sequestration of carbon by plantations growing in regions of low-rainfall in southern Australia. *For. Ecol. Manag.* **254**, 205–216 (2008).
67. Viera, M. & Rodríguez-Soalleiro, R. A complete assessment of carbon stocks in above and belowground biomass components of a hybrid eucalyptus plantation in southern Brazil. *Forests* **10**, 1–12 (2019).

68. Dagar, J. C. et al. Eucalyptus geometry in agroforestry on waterlogged saline soils influences plant and soil traits in North-West India. *Agric. Ecosyst. Environ.* **233**, 33–42 (2016).
69. Khaskhoussy, K., Kahlaoui, B., Misle, E. & Hachicha, M. Impact of irrigation with treated wastewater on physical-chemical properties of two soil types and corn plant (*Zea mays*). *J. Soil Sci. Plant Nutr.* **22**, 1377–1393 (2022).
70. Gelaye, Y. & Getahun, S. A review of the carbon sequestration potential of fruit trees and their implications for climate change mitigation: The case of Ethiopia. *Cogent Food Agric.* https://doi.org/10.1080/23311932.2023.2294544 (2024).
71. Jans, Y., Berndes, G., Heinke, J., Lucht, W. & Gerten, D. Biomass production in plantations: Land constraints increase dependency on irrigation water. *GCB Bioenergy* **10**, 628–644 (2018).
72. El Kateb, H., Zhang, H. & Abdallah, Z. Volume, biomass, carbon sequestration and potential of desert lands' afforestation irrigated by wastewater on examples of three species. *For. Ecol. Manag.* **504**, 119827. https://doi.org/10.1016/j.foreco.2021.119827 (2022).
73. Leonel, L. P. & Tonetti, A. L. Wastewater reuse for crop irrigation: Crop yield, soil and human health implications based on giardiasis epidemiology. *Sci. Total Environ.* **775**, 145833 (2021).
74. Gautam, S., Mishra, U., Scown, C. D. & Zhang, Y. Sorghum biomass production in the continental United States and its potential impacts on soil organic carbon and nitrous oxide emissions. *GCB Bioenergy* **12**, 878–890 (2020).
75. Yeboah, D., Burton, A. J., Storer, A. J. & Opuni-Frimpong, E. Variation in wood density and carbon content of tropical plantation tree species from Ghana. *New For.* **45**, 35–52 (2014).
76. Al-Shoaibi, A. Combined effects of salinity and temperature on germination, growth and gas exchange in two cultivars of Sorghum bicolor. *J. Taibah Univ. Sci.* **14**(1), 812–822. https://doi.org/10.1080/16583655.2020.1777800 (2020).
77. Iqbal, S., Farooq, M., Sanaullah, M., Ullah, A. & Nadeem, F. Effect of nitrogen application and sorghum mulch on nitrogen use efficiency, microbial biomass carbon, extracellular enzymes activities and growth of mashbean (*Vigna mungo* (L.) Hepper). *J. Plant Nutr.* **45**, 703–712 (2022).
78. Fernández, M. et al. Eucalyptus x urograndis biomass production for energy purposes exposed to a Mediterranean climate under different irrigation and fertilisation regimes. *Biomass Bioenerg.* **111**, 22–30 (2018).
79. Xuan, J. et al. Intelligent estimating the tree height in urban forests based on deep learning combined with a smartphone and a comparison with UAV-LiDAR. *Remote Sens.* **15**(1), 97. https://doi.org/10.3390/rs15010097 (2022).

Acknowledgements
The publication was financed by the Polish Minister of Science and Higher Education as part of the Strategy of the Poznan University of Life Sciences for 2024–2026 in the field of improving scientific research and development work in priority research areas.

Author contributions
A.S contributed on conceptualization, writing all the drafts of the manuscript and the final manuscript. Measuring, collecting, preparing, conserving, and transporting samples, analyzing samples results. T.I coordinated and supervised all the research work. M.D.B supervision and administrative assistance. J.B.D designed the research, assisted with samples and data analysis, revised the manuscript and gave suggestions, provided supervision. A.T provided reagents, helped in preparing conserving and analyzing samples. P.G contributed to performing sample analysis. M.A contributed to performing sample analysis. M.S.N helped with guidance for sample analysis.

Declarations

Competing interests
The authors declare no competing interests.

Additional information
Correspondence and requests for materials should be addressed to A.S.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.